

①a

$$G(s) = \frac{K e^{-\theta s}}{(1 + \tau_f s) \cancel{(1 + \tau_d s)}}$$

Stationær arvikeit:

$$\cancel{y_0} = 1,5 \quad u_1 = 1,7$$

$$\Downarrow \quad \Downarrow$$

$$y_0 = 34,25 \quad y_1 = 35,4$$

Her er $K = \frac{y_1 - y_0}{u_1 - u_0}$

$$\lim_{t \rightarrow \infty} \frac{y(t)}{t} = \dots = \boxed{5,75}$$

For $\downarrow$

$\lim_{t \rightarrow \infty} \frac{y(t)}{t} = \dots$

$$\hat{y} = G(s) \hat{u} \Rightarrow \Delta \hat{y} = G(s) \Delta \hat{u}$$

$$\Delta \hat{u} = \frac{\Delta u}{s}$$

der er konstant

$$\lim_{s \rightarrow 0} s G(s) \frac{\Delta u}{s} = \lim_{s \rightarrow 0} s G(s) \frac{\Delta u}{s}$$

$$= G(0) \Delta u = \frac{K e^{-\theta \cdot 0}}{(1 + \tau_f \cdot 0) \cancel{(1 + \tau_d \cdot 0)}} \Delta u = K \Delta u$$

$$\Downarrow$$

$$\frac{\Delta y}{\Delta u} = K$$

② e) det k er robust
da:

$$\operatorname{Re} \left(\frac{-k_2 \pm \sqrt{k_2^2 - 4 \cdot \overbrace{k_1}^{=0}}}{2} \right) < 0$$

$k_2 > 0 \Rightarrow -\frac{k_2}{2} < 0$

(2) e)

$$Y(s) = G(s) u(s) = \frac{1}{ms^2} u(s) \Rightarrow u(s) = ms^2 Y(s)$$

$$u(s) = -K y + F r = -k x_{est} + F_0 r$$

$$\dot{x}_{est} = A x_{est} + b u + L (y - x_{est})$$

$$y_{est} = c x_{est}$$

⇓

$$\dot{x}_{est} = (A - Lc - bk) x_{est} + b F_0 r + L y$$

$$u(s) = -K(s) y + F(s) r$$

$$G(s) = \frac{1}{ms^2}$$

$$K = k (sI - (A - Lc - bk))^{-1} L$$

$$F = (I - k (sI - (A - Lc - bk))^{-1} b) F_0$$

$$\underbrace{ms^2}_{G^{-1}} y = -K y + F r$$

$$y = \frac{F}{G^{-1} + K} r = \frac{GF}{1 + GK} r$$

$$\begin{aligned} F(s) &= \left(1 - (k_1, k_2) \left(\begin{pmatrix} s & 0 \\ 0 & s \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} k_1 & 0 \\ k_2 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ \frac{k_1}{m} & \frac{k_2}{m} \end{pmatrix} \right)^{-1} \begin{pmatrix} 0 \\ \frac{1}{m} \end{pmatrix} \right) F_0 \\ &= \left(1 - (k_1, k_2) \begin{pmatrix} s - L_1 & -1 \\ -L_2 - \frac{k_1}{m} & s - \frac{k_2}{m} \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ \frac{1}{m} \end{pmatrix} \right) F_0 = \left(1 - (k_1, k_2) \begin{pmatrix} \frac{1}{m} \\ s - L_1 - \frac{k_2}{m} \end{pmatrix} \right) F_0 \\ &= \left(1 - \frac{k_1}{m} + \frac{k_2 s}{m} - \frac{k_2^2}{m^2} \right) F_0 = \frac{s^2 - (L_1 + \frac{k_2}{m})s + \frac{k_2 L_2}{m} - L_2 - \frac{k_1}{m}}{s^2 - (L_1 + \frac{k_2}{m})s + \frac{k_2 L_2}{m} - L_2 - \frac{k_1}{m}} F_0 \end{aligned}$$

② ① $u = -kx + \rho_0 r = -(1.2)x + r$
 (2)

$\dot{x}_{01} = A x_{01} + b u + L (x - x_{01})$

$\dot{x}_{01} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} x_{01} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u + \begin{pmatrix} 0.2 \\ 0 \end{pmatrix} (x - x_{01})$

$\dot{x}_{01} = (A - L) x_{01} + (b + Lx) u$

$x = Cx = \begin{pmatrix} 1 & 0 \end{pmatrix} x$

③ $\dot{x} = Ax + bu$
 $y = Cx$

Først lit bok gjennom.

kanfor er p som har er 222

Her h du er i deep diving.
 (dette er ikke oppgaven (bare for eigen forfreming))

løsninga til systemet er p som er

$x(t) = e^{A(t-t_0)} x(t_0) + \int_{t_0}^t e^{A(t-\tau)} b u(\tau) d\tau$

vi vil sjå på styrbarhet altså om vi
 etter tid kan styre systemet slik vi vil.

Da må vi sjå på om det er slik at
 den vi styrer $\int_{t_0}^t e^{A(t-\tau)} b u(\tau) d\tau$ de kan

den kan bli, rett til alle tilstander
 har vi full styrbarhet men først ser vi dette?

Ja vi kan Taylor utvikle e bddt.

$\Rightarrow \int_{t_0}^t (I + A(t-\tau) + \frac{A^2(t-\tau)^2}{2!} + \dots) b u(\tau) d\tau$
 $= \int_{t_0}^t (b + A b (t-\tau) + \frac{A^2 b (t-\tau)^2}{2!} + \dots) u(\tau) d\tau$

vi ser at dette er en lineær kombi
 av $b, Ab, A^2b, \dots, A^{n-1}b$, men vi kan

vi kan sjå på det ikke, at ~~alle~~ alle

leddene er det, men er en lineær
 kombi av de første n leddene, derfor vil

~~alle~~ $[b \quad Ab \quad A^2b \quad \dots \quad A^{n-1}b]$ må ha rang som

er uavhengig e kolonne i $n \times n$ den er
 styrbart

② ①

$$Q = \begin{pmatrix} C \\ CA \\ CA^2 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix} \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{pmatrix}$$

$$C_1 = (1 \ 0 \ 0) \Rightarrow Q = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

~~Mat~~ Matrice er ikke
lineært uafhængig
denne er da ikke
små/bare

$$C_1 A = (-1 \ 0 \ 0)$$

$$C_2 A = (1 \ -1 \ 0)$$

$$C_3 A = (0 \ 1 \ -1)$$

$$C_1 A^2 = (1 \ 0 \ 0)$$

$$C_2 A^2 = (-2 \ 1 \ 0)$$

$$C_3 A^2 = (1 \ -2 \ 1)$$

$$C_2 = (0 \ 1 \ 0) \Rightarrow Q = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -1 & 0 \\ -2 & 1 & 0 \end{pmatrix}$$

ikke denne heller

$$C_3 = (0 \ 0 \ 1) \Rightarrow Q = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -2 & 1 \end{pmatrix}$$

her er matrice lineært
uafhængig, og da kan vi
vælge x_1 , x_2 og x_3

flycatcher
observer

$$\frac{da}{dr} = r^{4k} \cdot x^{(k)}$$

$$\begin{pmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{pmatrix}$$



$$(x_1, x_2, x_3)$$

$$\dot{x}_2 = v_1 - v_2 = x_1 - x_2$$

$$x_2' = v_2 - v_R = x_2 - x_1$$

endring $X_1 = -X_1$

$$\dot{X} = V, \quad X = X_1$$

sent add div
and the men
at far of
will fast in
X109
CPD

$$(s+1)^3 = 0$$

$$p_1 = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow p_1 = q$$

$$p_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow P_2 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{pmatrix}$$

X' bed dry

her kann es nicht sein
 x_1, e, x_2

2. $\sin^{-1} \frac{1}{2} = 30^\circ$ (acute angle)

$$A_{b_3} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$A^2 = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{pmatrix}$$

$$\begin{aligned} A^2 b_1 &= \begin{pmatrix} 1 \\ -2 \end{pmatrix} \\ A^2 b_2 &= \begin{pmatrix} 0 \\ -2 \end{pmatrix} \\ A^2 b_3 &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{aligned}$$

②

$$\begin{pmatrix} \ddot{p} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} p \\ v \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{m} \end{pmatrix} u$$

$$y = (1 \ 0) \begin{pmatrix} p \\ v \end{pmatrix}$$

u : er styrningskraft

p : position

v : speed

y : måling av pos fra satellittmålingssystem

$m = 1$

$\frac{1}{m} = 1$ - ønsket

③ Systemet er ustabilit da det har
to poler i 0

$$s^2 = 0 \quad \mathcal{L}^{-1}\left(\frac{1}{s^2}\right) = t$$

denne antas
her t antas

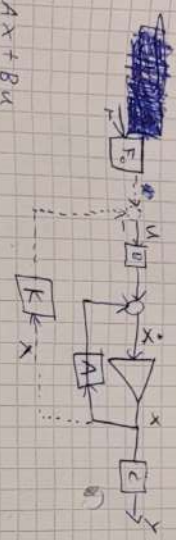
observerbarhet:

$$Q = \begin{bmatrix} p \\ cA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$cA = (1 \ 0) \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = (0 \ 0)$$

$\text{rang}(Q) = 1 \neq n = 2$ denne er
ikke fullstendig observerbar

2 b



$$\dot{x} = Ax + Bu$$

$$y = Cx$$

$$u = -Kx + F_0$$

$$\dot{x} = Ax + B(-Kx + F_0) + F_0$$

$$\dot{x} = (A - BK)x + BF_0 + F_0$$

$$y = Cx$$

$$A - BK = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} k_1 & k_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} k_1 & k_2 \\ k_2 & k_1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 \\ -k_2 & -k_1 \end{pmatrix}$$

$$s(s + \frac{k_2}{1}) + \frac{k_1}{1} = 0$$

$$s^2 + s \frac{k_2}{1} + \frac{k_1}{1} = 0$$

$$s = \frac{-\frac{k_2}{1} \pm \sqrt{(\frac{k_2}{1})^2 - 4 \frac{k_1}{1}}}{2} = -\frac{k_2}{2} \pm \frac{1}{2} \sqrt{k_2^2 - 4k_1}$$

$$s_1 = -1$$

$$s_2 = -1$$

$$k_2^2 - 4k_1 = 0$$

$$\begin{pmatrix} k_1 & k_2 \\ 1 & 4 \end{pmatrix}$$

$$k_1 = \frac{k_2^2}{4} = 1$$

$$K = \begin{pmatrix} 1 & 2 \end{pmatrix}$$

$$F_0 = \frac{1}{C(bK - A)^{-1}b}$$

$$\Rightarrow (bK - A)^{-1} = (A - bK)^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}^{-1} =$$

$$\begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & m \\ -1 & 0 \end{pmatrix}$$

$$C(bK - A)^{-1}b = (1 \ 0) \begin{pmatrix} 2 & m \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} =$$

$$= (1 \ 0) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1$$

$$F_0 = 1$$

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

$$\hat{x} = \frac{BF_0}{sI - (A - bK)} u$$

$$\hat{y} = C(sI - (A - bK))^{-1}bF_0 = (1 \ 0) \begin{pmatrix} s & -1 \\ 1 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ 1 \end{pmatrix} =$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}^{-1} =$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}^{-1} =$$

② b

$$\begin{pmatrix} 1 & 0 \\ 0 & s+2 \end{pmatrix} \begin{pmatrix} s+2 & 1 \\ -1 & s \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}{s(s+1)+1} = \frac{1}{s^2+s+1}$$

$$= \frac{1}{s^2+s+1} = \frac{1}{(s+\frac{1}{2}+i\frac{\sqrt{3}}{2})(s+\frac{1}{2}-i\frac{\sqrt{3}}{2})}$$

write as
stable

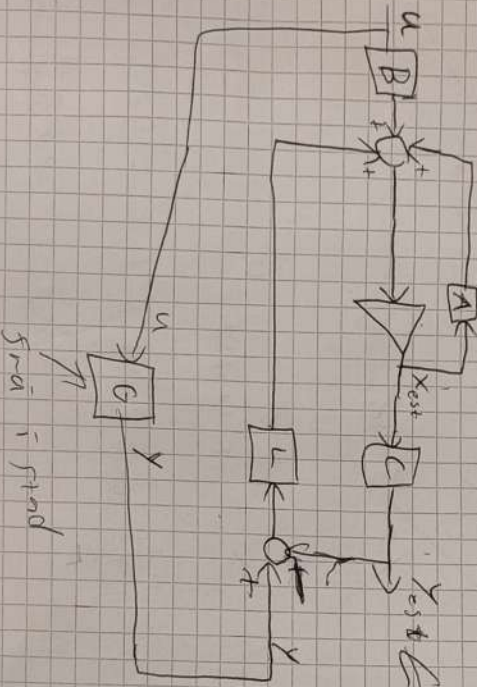
$$\frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$$

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s \hat{y}(s) = \lim_{s \rightarrow 0} s \frac{1}{s^2+s+1} = \hat{r} = 0$$

$$\lim_{s \rightarrow 0} \frac{1}{s^2+s+1} = \frac{1}{1} = 1 \quad \text{system det}$$

$$r = \text{stable} \quad \hat{y}(s) = \frac{1}{s^2+s+1} \quad \boxed{y \rightarrow 1}$$

2c



② c

$$\dot{x}_{err} = A x_{err} + b u + L (y - y_{err})$$

$$y_{err} = C x_{err}$$

$$y = Cx + w$$

$$\dot{x} = Ax + bu$$

$$\dot{x}_{err} = A x_{err} + b u + L (C x + w - C x_{err})$$

$$\dot{x}_{err} = A x_{err} - L C x_{err} + b u + L C x + L w$$

$$\dot{x}_{err} = (A - LC) x_{err} + b u + L C x + L w$$

$$\dot{e} = \dot{x} - x_{err} = Ax + bu - (A - LC)x_{err} - b u - L C x - L w = LCx - Lw$$

$$= (A - LC)(x - x_{err}) - Lw = (A - LC)e - Lw$$

$$A - LC = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} L_1 \\ L_2 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 0 & 1 \\ L_2 & 0 \end{pmatrix} = \begin{pmatrix} -L_1 & 1 \\ -L_2 & 0 \end{pmatrix}$$

$$A_e = \begin{pmatrix} -L_1 & 1 \\ -L_2 & 0 \end{pmatrix}$$

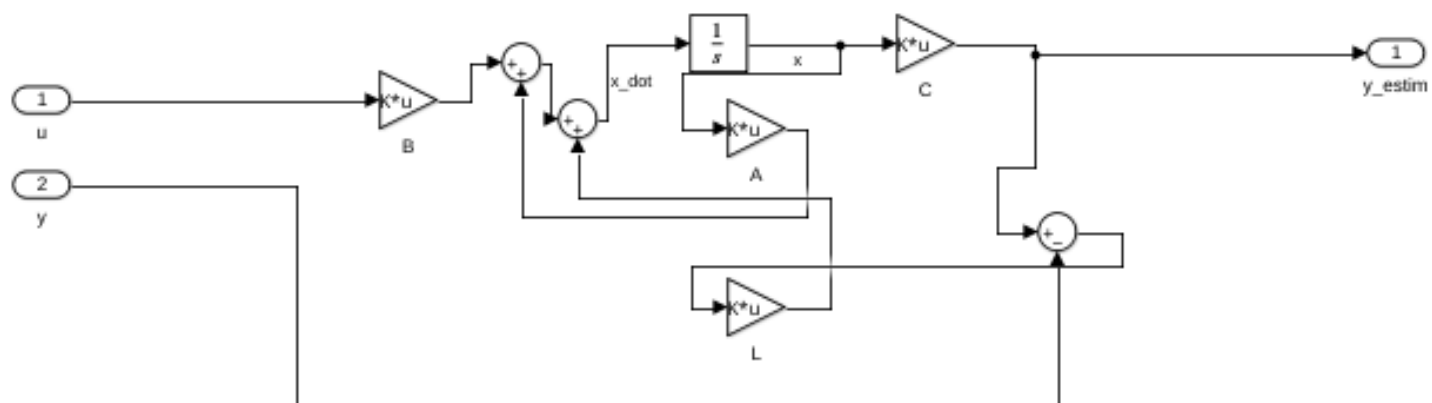
$$s^2 - 10 = 0$$

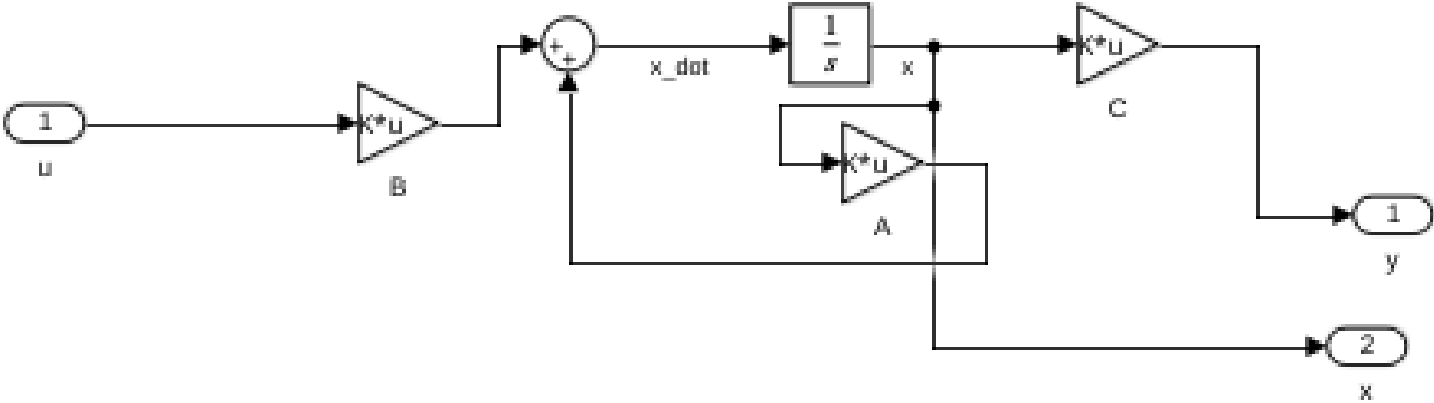
$$(s+L_1)s + L_2 = 0$$

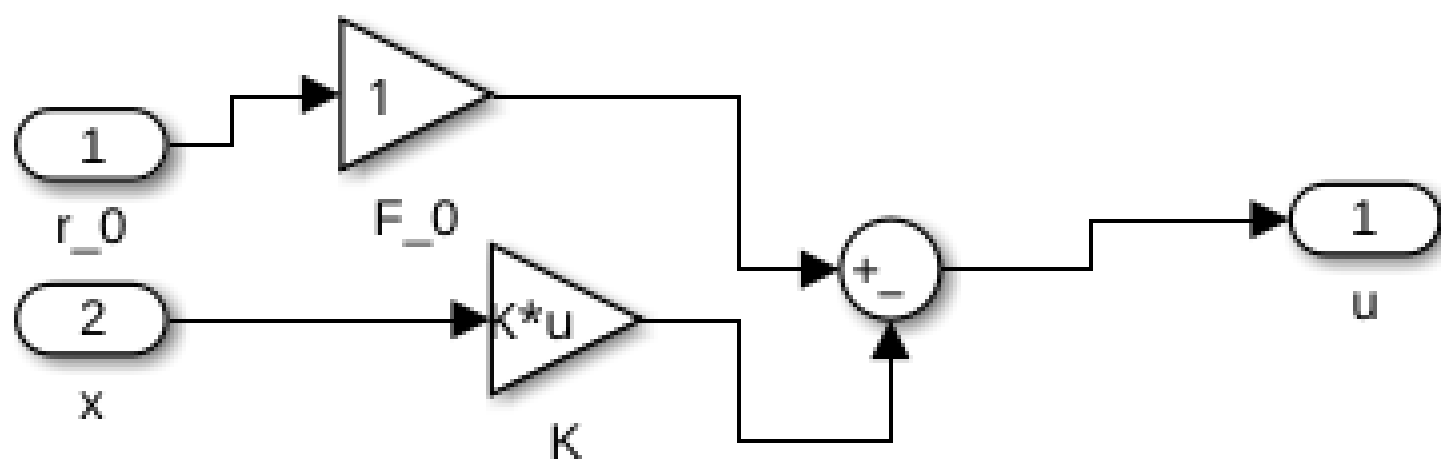
$$s^2 + L_1 s + L_2 = 0$$

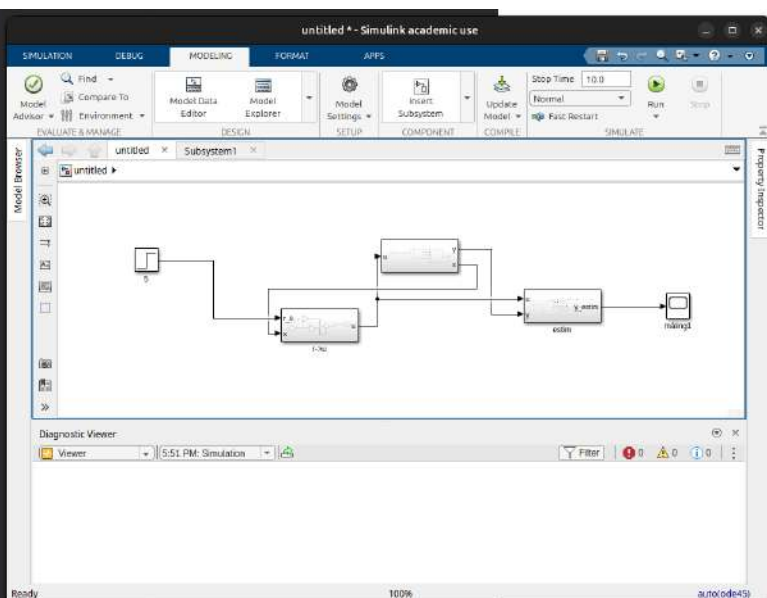
$$L = \begin{bmatrix} +20 \\ 100 \end{bmatrix}$$

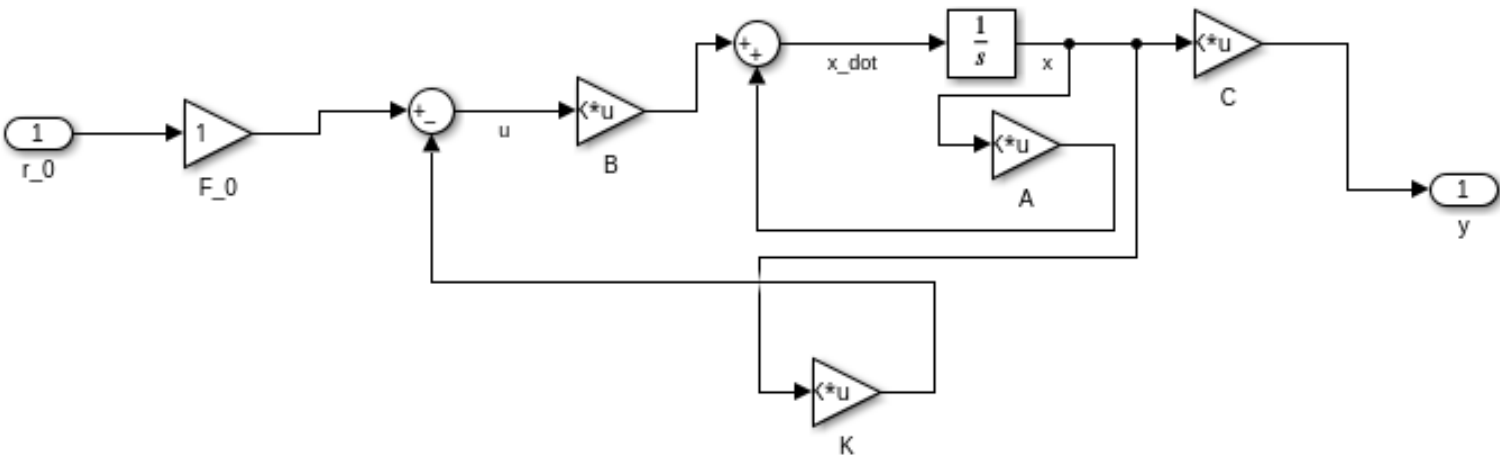
$$\begin{aligned} L_2 &= \frac{L_1^2 - 4L_2}{2} \\ L_2 &= \frac{400 - 400}{2} = 0 \\ L_2 &= \frac{4 \cdot 100 - 100}{4} = 100 \end{aligned}$$

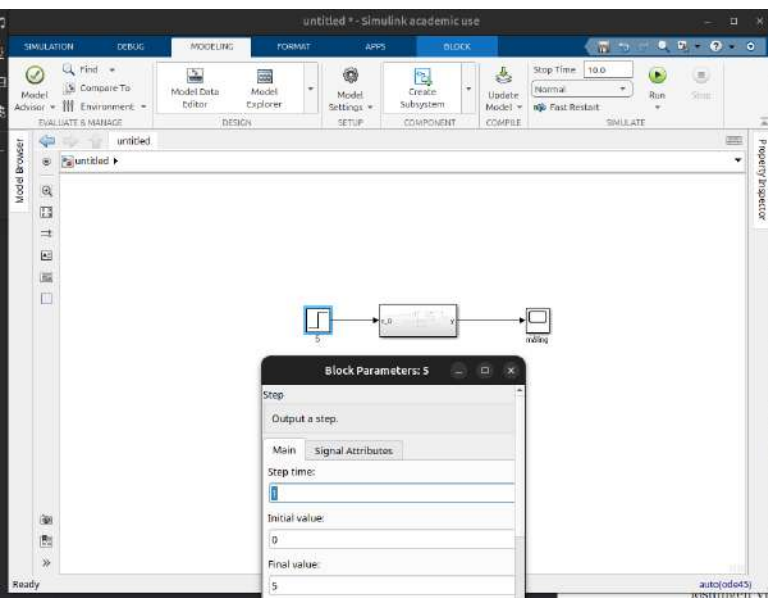












$$F_0 = \frac{1}{c(bk - A)^{-1}b} \quad (4)$$

$y_{sc} = r_0$ ved likevekt.

tilbakekobling

timerings

ningene

parameter

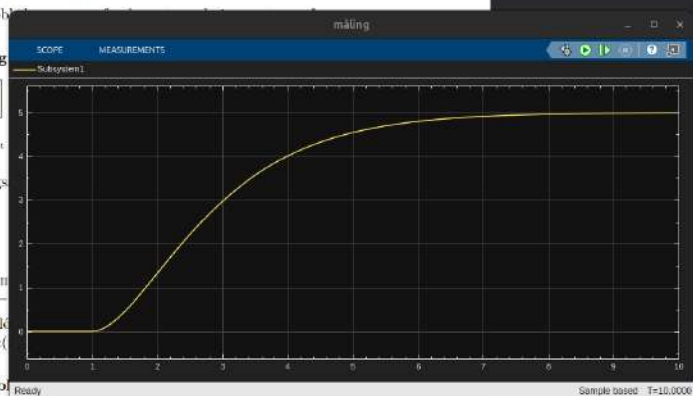
er $\lambda_1 = -$

og simul

pådrag $u(t)$

skt.

bakekob



$$u = - \begin{bmatrix} k_1 & k_2 \end{bmatrix} \begin{bmatrix} p_{est} \\ v_{est} \end{bmatrix} + F_0 r. \quad (8)$$

entene k_1 , k_2 og F_0 ble beregnet i oppgave b) og algoritmen for å estimatene p_{est} og v_{est} ble utviklet i oppgave c).

emet (system + tilbakekobling + estimator) og vurder hvorvidt løsningen virker.